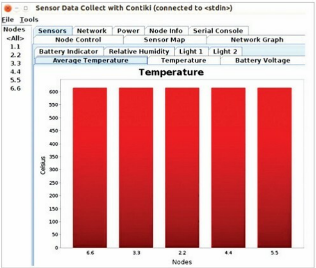

Figure 8: Temperature evaluation in Celsius at the motes

Installation of Foren6 with Contiki

To configure the dependencies, type:

```
$ sudo apt-get install -y qt4-qmake libqt4-dev make
libexpat1-dev cmake libpcap0.8-dev libc6-dev tshark gcc g++
```

To get the source code with GIT, type:

```
$ git clone https://github.com/cetic/foren6.git
```

Use the following commands to install Foren6:

```
$ cd foren6
$ make
$ sudo make install
```

Configuration of Sniffer to activate real-time capturing

To access and work on USB serial devices, the Linux Group 'dialout' is used. Initially, Foren6 is launched as the root user. If any other user account is used, then the user should be assigned the Linux Group 'dialout' to use USB.

```
$ sudo adduser <username> dialout
Sniffer Programming
```

Figure 9: Foren6 as a 6LoWPAN analyser

```
$ git clone https://github.com/cetic/contiki
$ cd contiki
$ git checkout sniffer
$ make TARGET=sky savetarget sniffer.upload
```

Scope of IoT based research simulations using Cooja

The Internet of Things is an emerging domain of research at various academic as well as corporate establishments. Because of the increasing number of devices currently connected under the IoT, there are vast areas for research in this domain.

The following are some of the approaches with which novel and effectual algorithms can be devised and implemented using Cooja:

- Interoperability and cross-protocol compatibility
- Development of energy-aware IoT scenarios
- Power aware scheduling and routing
- Prediction and avoidance of energy consumption attacks
- Lifetime analytics for robust IoT environments
- Reproducible and multi-interface implementations END

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